CSCE604227 System Programming
CSCE604227 Pemrograman Sistem
Week 00: Overview

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Sendirian Berhad

https://sp.vlsm.org/Slides/sp00.pdf
Always check for the latest revision!

REV021 06-Feb-2023

SP221[1]): System Progamming

Week	Topic
Week 00	Overview
Week 01	Linux Kernel and Programming Interface
Week 02	Revisit Linux From Scratch
Week 03	FUSE: Filesystem in Userspace
Week 04	Project FUSE 1
Week 05	Project FUSE 2
Week 06	Project FUSE 3
Week 07	Project FUSE 4
Week 08	Project FUSE 5
Week 09	Project FUSE 6
Week 10	Project FUSE Presentation

[1]) This information will be on **EVERY** page two (2) of this course material.

STARTING POINT — https://sp.vlsm.org/

- **Text Book** — The Linux Programming Interface, 2010, No Starch Press, ISBN 978-1-59327-220-3 — https://man7.org/tlpi/.
- **Resources**
  - **SCELE** — https://scele.cs.ui.ac.id/course/view.php?id=3545. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** https://github.com/os2xx/sysprog/: sp00.pdf (W00), sp01.pdf (W01), sp02.pdf (W02), sp03.pdf (W03), sp04.pdf (W04), sp05.pdf (W05), sp06.pdf (W06), sp07.pdf (W07), sp08.pdf (W08), sp09.pdf (W09), sp10.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **This is How Me DO IT!** — https://doit.vlsm.org/
    - PS: "Me" rhymes better than "I" duh!

Agenda

1. Start
2. Schedule
3. Agenda
4. How to contact the Lecturer
5. Assessment
6. Final Grade
7. The Three-Strikes Rule
8. Assignments
9. This is an elective course!
10. Miscellaneous

How to contact the Lecturer

- **Always introduce yourself**. State your "`GitHubAccount`", "`Student ID`", "`Hypervisor`", and "`SP class`".
- Post a question/query on **SCELE SP231** — (The enrollment key is **XXX**): https://scele.cs.ui.ac.id/course/view.php?id=3545.
- For SIAK related questions, use email: (Subject:**[SP]**) `rms46(AT)ui.ac.id`.
- **DO NOT** send an email for assignment-related questions.

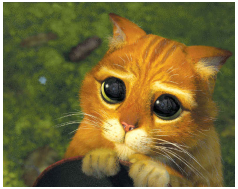

Figure: Never ever whine and pretend like this[1]!

[1] "Puss in Boot" is a DreamWorks/Paramount Picture character.

Emails between a "Gen Z" and a "Babyboomer"

selamat malam pak,

saya cicak salah satu mahasiswa sistem operasi di kelas bapak,
dengan username : cbkadal

1 saya ingin bertanya, kenapa XXXXX YYYYY ZZZZZ ya pak??

XXXXX YYYYY ZZZZZ, kalau boleh tau kesalahan saya dimana ya pak??

untuk 2 atiannya saya ucapkan terima kasih.

Salam,

3

Cicak Bin Kadal

Hallo Gen Z Zaman Now!

1 Kalimat baru seharusnya selalu dimulai dengan huruf besar.

2 Tanda baca seperti ":" (titik dua) seharusnya tanpa spasi. "Ini betul: ", "Ini salah : ".

3 Mengapa sampai lebih dari satu tanda-tanya??????????????

Salam,

Babyboomer.

Assessment

- **11 Weekly Assignments @ 11.11 points**.
  - Assignments will vary from week to week.
  - The assignment deadline will be by the end of every week. See https://sp.vlsm.org/#idx02.
  - Check your points regularly at https://academic.ui.ac.id/
  - See also, https://sp.vlsm.org/Log/.
  - **DO NOT COMPLAIN** weeks after!
- You need to log your weekly activities!
  - See https://osp4diss.vlsm.org/ETC/logCodes.txt
  - See https://cbkadal.github.io/sp231/TXT/mylog.txt
  - **3 SKS** (Units) means 9 hours (540 minutes) per week!

Final Grade (1)

- The final grade will be the best 9 out of 11 assignments.
- Do not ask for any dispensations like a broken computer, circumcision (sunat), cold, competitions (including Gemastik), deadline extension, influenza, lame excuses, marriage, mourning, power failure, remedial, return to the village (mudik), slow network (lemot), two-semester evaluation, umrah, weddings, etc.
- It also includes: "It is not my fault but of $\{X : X \in \textit{Lecturer} \parallel \textit{Fasilkom} \parallel \textit{UI} \parallel \textit{Kampus Merdeka} \parallel \textit{Immigration} \parallel \textit{Foreign Embassy} \parallel \textit{else}\}$."
- **Two (2) "spare" assignments will be more than enough!**
- In case of emergency, contact your Academic Advisor!

Final Grade (2)

- C-2C (C minus to C)
  - Up to 5 points, only if:
    - your grade is between 50.00 and 55.00, and
    - you have a "good" track record.
- Score Range

85 - ... = A	80 - 85 = A-	75 - 80 = B+	70 - 75 = B
65 - 70 = B-	60 - 65 = C+	55 - 60 = C	50 - 55 = D or C[1]
40 - 50 = D	30 - 40 = E	20 - 30 = E	00 - 20 = E

[1] C-2C: terms and conditions apply — void where prohibited by law.

Grade Examples

Mata Kuliah: CSCM602055 - Sistem Operasi
Kelas: 696070 - Sistem Operasi A, B, C, INT
[redacted]

-1 = no exam

Nama	NPM	W00	W01	W02	W03	W04	W05	W06	W07	W08	W09	W10	UTS	UAS	C-2C	TOTAL	
1234567890	1234567890	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	0	99.99	A
1234567891	1234567891	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	-1	-1	0	99.99	A
1234567892	1234567892	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	0	0	0	99.99	A
1234567893	1234567893	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	0	0	-1	-1	0	99.99	A
1234567894	1234567894	0	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	11.11	0	-1	-1	0	99.99	A
1234567895	1234567895	6.01	6.01	6.01	6.01	8.31	6.01	5.55	10.55	5.55	5.55	5.55	-1	-1	0	60.01	C+
1234567896	1234567896	6.01	6.01	6.01	6.01	8.31	6.01	5.55	10.55	5.55	5.55	5.55	0	0	0	59.55	C
1234567897	1234567897	6.01	6.01	6.01	6.01	8.31	6.01	5.55	10.55	5.55	5.55	5.55	11.1	11.1	0	70.65	B
1234567898	1234567898	6.21	4.88	6.21	3.11	6.21	6.21	6.21	6.01	6.01	6.01	6.01	-1	-1	0	55.09	C
1234567899	1234567899	6.21	4.88	6.21	3.11	6.21	6.21	6.21	6.01	6.01	6.01	6.01	0	0	1.05	55.01	C
1234567900	1234567900	6.21	4.88	6.21	3.11	6.21	6.21	6.21	6.01	6.01	6.01	6.01	11.1	11.1	0	65.27	B-

The Three-Strikes Rule

- All major academic rules violations will be handled directly by the Faculty of Computer Science, University of Indonesia.
- "Accidents" may happen. There will be warnings for the first two minor violations.
- Your final grade will be reduced for the third warning.
- Your final grade will be reduced to "D" for the fourth warning.
- Five (5) or more warnings will be considered as a significant academic-rules violation.

Assignments

- There will be no mid-term (UTS) nor final-term (UAS). Instead, there will be 11 weekly assignments. Your grade will be taken from the best 9 out of 11 assignments.
- You need to run "VirtualBox" on a computer with more than 4GB RAM and up to 64 GB disk space.
- Each assignment deadline will be by the end of that "week". The weekly schedule will be at https://sp.vlsm.org/#idx02.
- Submit (push) the assignments to https://github.com/. If you still don't have one, you need to sign up for a https://github.com/ account. More information will follow.
- See the assignment list at https://osp4diss.vlsm.org/ASP.html.

This is an elective course!

- You are not required to take this course!
- This course is not for you if you:
  - do not like the Operating Systems course.
  - do not like to get your hands dirty.
  - do not have enthusiasm nor initiative at all.
  - do not like to "hack".
- Cold Feet? Second Guess? You might want to drop this course now (this week)!
- This is the way!

Out of Topic/Intermezzo/Segue

- Semiconductor Scalling:
  - Process Shrink: $10\mu m$ (1971), $250nm$ (1996), $10nm$ (2016), $5nm$ (2020), $3nm$ (2022).
  - Smaller Devices means:
    - Less space.
    - Less power consumption.
    - More density.
- Indonesia:
  - Fairchild Semiconductor Indonesia.
  - National Semiconductor Indonesia.
  - Minister of Manpower (Menteri Tenaga Kerja) 1983–1988.
- Technology:
  - SoC: System on a Chip.
  - SiP: System in a Package.
  - Fab/Foundry: Taiwan Semiconductor Manufacturing Company (TSMC), Ltd.
    - Have No Fab? It is OK! E.g., Marvell Technology, Inc (1995).
  - Lithography: ASML Holding, N.V: Advanced Semiconductor Materials Lithography.
  - Optics: Carl Zeiss SMT GmbH (This is NOT Optik Seis, Duh :).

TSMC Logic Nodes

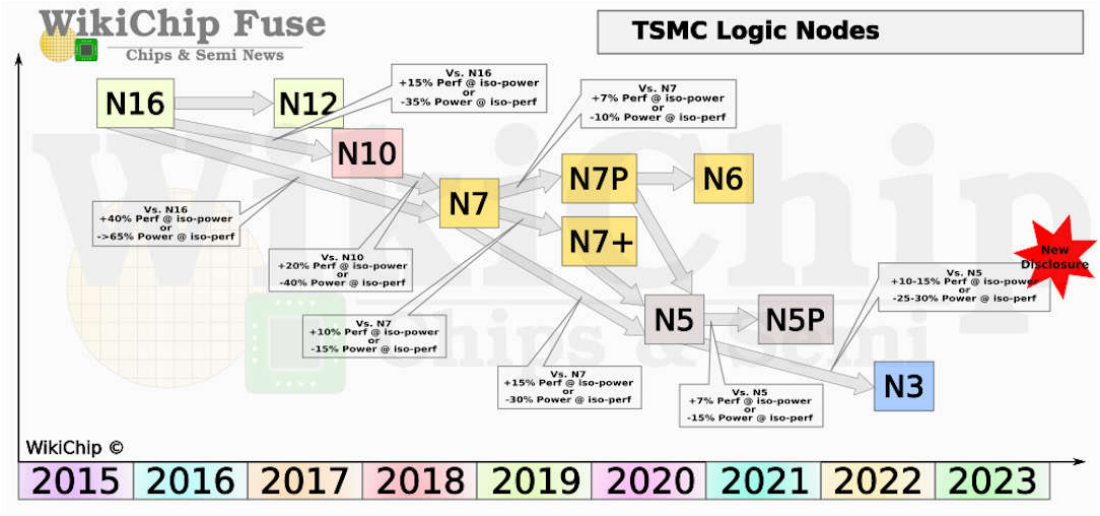

Figure: Source: WikiChip

The Computing Diciplines

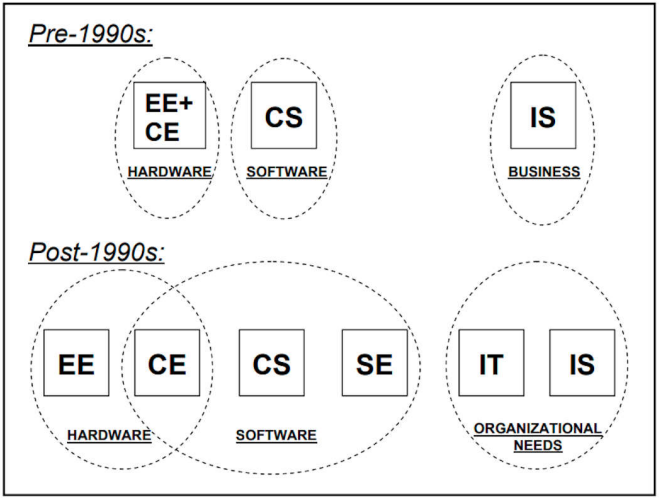

Figure 2.1. Harder Choices: How the Disciplines Might Appear to Prospective Students

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